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Digital Systems Testing and Verification

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SPI Master Controller Verification

Test Plan

DUT: APB-slave SPI Master IP (8/16/32-bit, 4 SS, 8-deep FIFOs)

Methodology: SystemVerilog-only verification environment

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1. Introduction

This test plan defines the verification approach for the final SPI Master Controller project. The environment verifies the APB register interface, SPI transfer protocol, FIFOs, interrupts, loopback, programmable clock divider and inter-transfer delay behavior. The document is intentionally separated from the final report: this file describes what is planned and how it is checked, while the final report records actual bug findings, coverage numbers and lessons learned.

- Verification objective: detect functional deviations from the SPI Master specification using self-checking tests, a reference model, functional coverage and SVA.
- Methodology: SystemVerilog-only verification with APB and SPI BFM, directed tests, randomized stimulus classes and requirement-driven checkers.
- Primary grading goal: pass mandatory tests, reach coverage targets and flag faulty DUT behavior with recognized log strings.

2. Scope, References and Assumptions

The test plan covers the public verification scope and does not rely on the staff-only faulty RTL or bug catalog. All planned checks are derived from the public SPI Master specification and the grading contract.

Reference	Usage in Test Plan
SPI_Master_Spec.pdf	Source of DUT behavior, register map, SPI timing, FIFO rules, interrupt semantics and requirements R1–R25.
grading_interface.md	Source of mandatory test names, log-message contract, Makefile/regression expectations and coverage gate.
README.md	Source of deliverables: docs/test_plan.pdf with covergroup + SVA inventory; docs/final_report.pdf with bugs, coverage numbers and lessons learned.
Final SV project files	Actual implementation basis: tb_top.sv, APB/SPI BFM, mandatory tests, coverage.sv, ref_model.sv, stim.sv, spi_sva.sv, Makefile and run_parallel.py.

Assumptions

- The verification plan is based on the public SPI Master specification rather than copied behavior from the golden RTL.
- The staff-only faulty RTL, bug catalog and reference environment are unavailable to students; all checkers are requirement-driven.

- The internal hierarchy `tb_top.u_wrap.u_dut.u_regfile` and `tb_top.u_wrap.u_dut.u_core` is used for SVA binding.
- Randomized tests are deterministic through +SEED, and total regression invocations remain below the grading limit.
- Coverage goals are planned targets; actual coverage percentages and missed bins are reported in `final_report.pdf` and `coverage_report.pdf`.

3. DUT Features Under Verification

- APB v2.0 zero-wait-state register access with 32-bit aligned offsets.
- CTRL, STATUS, TX_DATA, RX_DATA, CLK_DIV, SS_CTRL, INT_EN, INT_STAT and DELAY programmer model.
- SPI modes 0, 1, 2 and 3 using CPOL and CPHA.
- 8-bit, 16-bit and 32-bit transfer widths.
- MSB-first and LSB-first shifting.
- Programmable CLK_DIV including DIV=0 and large divider values.
- Programmable DELAY between consecutive transfers.
- TX and RX FIFOs with depth 8, full, empty and overflow behavior.
- Five sticky W1C interrupt sources and combined IRQ equation.
- Loopback mode and software-controlled active-low slave-select outputs.

4. Verification Environment Architecture

The environment is organized around a task-based SV-only testbench. `tb_top.sv` instantiates the DUT wrapper, common BFM, the reference model and coverage collector, then dispatches tests based on +TESTNAME or +UVM_TESTNAME. The all option executes the implemented test list sequentially with reference-model reset between tests.

Component	Role in Verification Environment
<code>tb_top.sv</code>	Instantiates PCLK/PRESETn, APB/SPI interfaces, <code>dut_wrapper u_wrap</code> , APB BFM, SPI slave BFM, reference model, coverage collector, test dispatcher, timeout and SVA binds.
<code>apb_master_bfm.sv</code>	Task-based APB master for zero-wait-state register reads and writes.
<code>spi_slave_bfm.sv</code>	SPI slave responder/monitor controlled by <code>tb_top</code> signals: mode, width, <code>lsb_first</code> and MISO pattern.
<code>ref_model.sv</code>	Reference model for register mirrors, FIFO queues, interrupt state, expected IRQ, transfer prediction and <code>error_count</code> .

coverage.sv	Functional coverage collector with covergroups for configuration, divider, interrupts, delay, slave select, loopback, FIFO depth, busy, transfer done, register access and reset.
stim.sv	Randomizable stimulus classes: spi_txn, reg_acc_txn, irq_txn, error_txn, fifo_stress_txn, clk_div_corner_txn and delay_txn.
spi_sva.sv	SVA modules bound to the register file and SPI core to check APB, IRQ, FIFO and SPI timing/protocol properties.
tests/*.sv	Ten mandatory tests with scenario-specific directed and randomized checks.

Compile/run flow: APB/SPI interfaces + DUT RTL + BFM's + ref_model/coverage/stim + tests + SVA + tb_top.

Main pass/fail contract: [TEST_PASSED] <test_name> or [TEST_FAILED] <test_name> errors=<n>. Bug-detection messages: [SCOREBOARD_ERROR], [ASSERTION_ERROR], [CHECKER_ERROR].

5. Test Strategy

- Directed tests are used for reset values, register access, FIFO boundaries, interrupt W1C, divider corners, loopback, delay and error conditions.
- Randomized transaction classes in stim.sv extend coverage for configuration values, FIFO bursts, interrupt data, error values and delay ranges.
- The reference model predicts register values, FIFO queues, interrupt status, IRQ and RX/TX data; mismatches print [SCOREBOARD_ERROR].
- SVA checks APB protocol, zero-wait-state behavior, interrupt equation, FIFO overflow conditions and SPI timing/protocol properties.
- Functional coverage is sampled explicitly by tests through spi_coverage_col sample_* tasks.

6. Mandatory Test Inventory

Test Name	Purpose	Main Planned Checks / Stimulus	Reqs
sanity_test	Smoke / basic transfer	Mode 0, 8-bit, MSB-first transfer; checks RX_DATA, INT_STAT[TRANSFER_DONE], TX_EMPTY, IRQ assertion and W1C clear.	R4, R6, R7, R9, R10, R16, R17
reg_access_test	Register access and reset	Reads reset values; performs read/write checks on R/W registers; checks STATUS, TX_DATA, RX_DATA behavior and W1C behavior.	R1, R2, R15, R17, R22, R23

mode_coverage_test	Mode / bit-order / width coverage	Runs SPI mode, width and bit-order combinations with reference-model checks and coverage sampling.	R4, R5, R6, R7, R19, R25
width_coverage_test	Width boundaries	Runs 8-, 16- and 32-bit transfers; verifies width masking and RX data; samples configuration, loopback, FIFO count, busy, transfer-done and register coverage.	R6, R7, R9, R10
fifo_stress_test	FIFO depth and overflow stress	Random legal bursts; RX_FULL after 8 entries; TX_FULL/TX_OVF; W1C behavior; RX drain; empty RX read; FIFO count coverage.	R9, R10, R11, R12, R13, R14, R15, R17
interrupt_test	Interrupts, masking, W1C, race	Targets TRANSFER_DONE, TX_EMPTY, TX_OVF, RX_FULL, RX_OVF, masked status capture, multi-bit W1C and W1C/event race stimulus.	R13, R14, R16, R17, R18
clk_div_corner_test	Clock divider corners and latching	Measures SCLK period for DIV corner values; verifies DIV=0 behavior, idle polarity, liveness and mid-transfer divider latch behavior.	R4, R7, R8, R24, R25
loopback_test	Loopback self-test	Runs loopback enabled/disabled across widths; verifies RX equals TX when loopback is enabled and external MISO is ignored.	R6, R7, R10, R19
delay_transfer_test	Inter-transfer delay	Directed delay x divider sweep; BUSY held during large delay gap; mid-transfer CLK_DIV write; randomized delay sweep; RX checks after multi-word bursts.	R21, R7, R10, R25
error_injection_test	Illegal/reserved/error cases	Reserved and non-aligned addresses; illegal WIDTH; extreme CLK_DIV/DELAY; all SS patterns; empty RX read; W1C; W1C race.	R13, R15, R17, R18, R20, R23, R24

7. Requirement Traceability Matrix

The matrix maps each numbered requirement to planned tests, checker responsibility, coverage and SVA. It is the central traceability artifact for docs/test_plan.pdf.

R1-R13

Req	Requirement Summary	Tests	Checker / Scoreboard	Coverage	SVA / Assertion
R1	R/W registers return last written value masked by RO bits.	reg_access_test, error_injection_test	Reference model mirrors CTRL/CLK_DIV/SS_CTRL/INT_EN/DELAY and compares APB reads.	cg_reg_writen, cg_reg_read	APB timing SVA supports legal access.

R2	Registers return reset values after PRESETn.	reg_access_test	Reset-value reads and reference-model reset.	cg_reset_values	IRQ_4_INT_STAT_ZERO_AFTER_RESET.
R3	CTRL.EN=0 holds shifter/FIFOs reset; SCLK idle; SS_n high.	reg_access_test, clk_div_corner_test	Disable/flush checks and SCLK idle checks.	cg_busy, cg_ss_select	SPI_1_SCLK_IDLE_POLARITY.
R4	SCLK idle polarity matches CPOL.	mode_coverage_test, clk_div_corner_test	check_sclk_idle and SVA.	cg_config	SPI_1_SCLK_IDLE_POLARITY.
R5	MOSI stable across sample edge.	mode_coverage_test	SPI BFM/monitor and sample-edge assertions.	cg_config cross	SPI_2A/B_MOSI_STABLE_SAMPLE.
R6	MSB/LSB-first shift order.	mode_coverage_test, width_coverage_test	Reference model and RX compare.	cg_config cross	Data checked by scoreboard/BFM.
R7	Transfer length and BUSY behavior.	sanity_test, width_coverage_test, clk_div_corner_test, delay_transfer_test	STATUS polling, liveness and transfer_done checks.	cg_busy, cg_transfer_done	SPI_5, SPI_6.
R8	SCLK frequency equals $PCLK/(2*(DIV+1))$.	clk_div_corner_test	Measured SCLK period against expected.	cg_clk_div	Timing checker.
R9	TX writes accepted while not full and FIFO order maintained.	sanity_test, fifo_stress_test	TX FIFO model and burst checks.	cg_tx_fifo, cg_reg_writen	SPI_5_TX_POP_ONE_CYCLE_PULSE.

R10	RX reads pop FIFO in order.	width_cover age_test, fifo_stress_t est, loopback_te st, delay_transf er_test	RX_DATA readback compared to reference model.	cg_rx_fifo, cg_reg_read	Scoreboard/checker.
R11	TX FIFO depth exactly 8; TX_FULL on 8th pending entry.	fifo_stress_t est	TX_FULL/TX_EMPTY status checks after 8 writes.	cg_tx_fifo	Scoreboard/FIFO coverage primary; IRQ_1 supports overflow side effect.
R12	RX FIFO depth exactly 8; RX_FULL on 8th entry.	fifo_stress_t est, interrupt_te st	RX_FULL status/interrupt and drain checks.	cg_rx_fifo, cg_interrupt_s ources	FIFO_2_RX_COUNT_BO UNDED.
R13	TX overflow discards write and sets TX_OVF.	fifo_stress_t est, interrupt_te st, error_injecti on_test	TX overflow stimulus and status/INT_STAT checks.	cg_interrupt_s ources, cg_tx_fifo	IRQ_1_TX_OVF_SET_ON_ OVERFLOW.

R14-R25

Req	Requirement Summary	Tests	Checker / Scoreboard	Coverage	SVA / Assertion
R14	RX overflow discards word and sets RX_OVF.	fifo_stress_test, interrupt_test	RX overflow stimulus and INT_STAT checks.	cg_interrupt_sour ces, cg_rx_fifo	FIFO_3_RX_OVF_SET_ON_OVERF LOW.
R15	Empty RX read returns zero and does not set RX_OVF.	fifo_stress_test, error_injection_ test, reg_access_test	Empty RX read check.	cg_rx_fifo, cg_reg_read	Scoreboard/checker.
R16	IRQ = (INT_STAT & INT_EN).	sanity_test, interrupt_test	IRQ read/check and masking scenarios.	cg_irq_masked, cg_interrupt_sour ces	IRQ_2_IRQ_EQUATION.

R17	INT_STAT is W1C.	reg_access_test, fifo_stress_test, interrupt_test, error_injection_test	Write-0 no clear, write-1 clear checks.	cg_irq_w1c	IRQ_3 sticky checks.
R18	W1C race event wins.	interrupt_test, error_injection_test	Race scenario checks DONE remains captured.	cg_irq_w1c, cg_interrupt_sources	Checker targeted.
R19	Loopback routes MOSI to RX and ignores MISO.	loopback_test, mode_coverage_test, delay_transfer_test	RX equals TX when LOOPBACK =1.	cg_loopback	Scoreboard/checker.
R20	SS_n follows SS_CTRL; no autonomous toggling.	error_injection_test, fifo_stress_test	SS pattern coverage and transfer precondition checks.	cg_ss_select	SPI_3 partially supports transfer precondition; SS_CTRL formula checked by scoreboard/test.
R21	DELAY idle half-cycles between consecutive transfers.	delay_transfer_test	Directed delay/div sweep, BUSY gap check and randomized delay sweep.	cg_delay	Delay checker/test.
R22	APB zero-wait-state and no slave error.	all APB-based tests	APB BFM and SVA.	cg_reg_read/write	APB_1, APB_2, APB_3a/3b, APB_5, APB_6.
R23	Reserved offsets read zero and writes ignored.	error_injection_test, reg_access_test	Reserved and non-aligned address checks.	cg_reg_read via tests	APB_4a_PRDATA_RESERVED_ZERO.
R24	CLK_DIV=0 gives PCLK/2.	clk_div_corner_test, error_injection_test	DIV=0 period check and coverage.	cg_clk_div div_zero	Timing checker.

R25	DIV, MODE, WIDTH, LSB_FIRST sampled at transfer start.	clk_div_corner_test, mode_coverage_test, delay_transfer_test	Mid-transfer CLK_DIV change and latched-mode sample-edge assertions.	cg_config, cg_clk_div	SPI_2 uses xfer_mode.
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8. Covergroup Inventory

coverage.sv implements a class-based functional coverage collector. Tests call sample_* tasks explicitly when stimulus or observed behavior reaches a planned coverage point. The minimum planned targets include 4 SPI modes x 3 legal widths x 2 bit orders, CLK_DIV corners 0/1/2/3/255/1024/65535, FIFO occupancy empty/1/mid/7/full, all five interrupt sources asserted/cleared/masked, register read/write/reset observations, DELAY 0/1/large and loopback per width.

Covergroup	Coverpoints / Bins	Purpose	Requirements
cg_config	mode, lsb_first, width; cross mode x width x order; illegal width sampled separately as negative stimulus.	Covers all four modes, MSB/LSB-first and legal 8/16/32-bit widths; WIDTH=2b11 is negative/error stimulus.	R4, R5, R6, R7, R25
cg_clk_div	div_zero, div_one, div_small, div_typical, div_medium, div_255, div_large, div_1024, div_very_big, div_max.	Covers divider corners and ranges including 0, 1, 255, 1024 and 65535.	R8, R24, R25
cg_interrupt_sources	TX_EMPTY, RX_FULL, TX_OVF, RX_OVF, DONE fired bins.	Ensures every interrupt source is stimulated.	R13, R14, R16
cg_irq_w1c	Clear transitions for all five INT_STAT bits.	Covers W1C clearing of sticky interrupt bits.	R17, R18
cg_irq_masked	{INT_STAT bit, INT_EN bit} for each source.	Covers masked/unmasked status capture and IRQ gating.	R16
cg_delay	delay_zero, delay_min, delay_small, delay_medium, delay_large.	Covers DELAY=0, DELAY=1 and large values up to 255.	R21
cg_ss_select	slave0_only, slave1_only, slave2_only, slave3_only, no_slave, multi_slave.	Covers single, none and multi-slave select patterns.	R20
cg_loopback	loopback on/off x width.	Covers loopback and non-loopback transfers per width.	R19

cg_tx_fifo	empty, one, partial, mid, almost full, full, empty_to_full, full_to_empty.	Covers TX FIFO occupancy boundaries and transitions.	R9, R11, R13
cg_rx_fifo	empty, one, partial, almost full, full, empty_to_full, full_to_empty.	Covers RX FIFO occupancy boundaries and transitions.	R10, R12, R14, R15
cg_busy	idle and active bins.	Covers BUSY status states.	R7
cg_transfer_done	done bin.	Covers completion event.	R7, R16
cg_reg_written	CTRL, TX_DATA, CLK_DIV, SS_CTRL, INT_EN, INT_STAT, DELAY.	Covers writes to implemented writable register ports.	R1, R9, R17, R22
cg_reg_read	CTRL, STATUS, RX_DATA, CLK_DIV, SS_CTRL, INT_EN, INT_STAT, DELAY.	Covers reads of implemented readable register ports.	R1, R10, R15, R22
cg_reset_values	reset reads for CTRL, STATUS, CLK_DIV, SS_CTRL, INT_EN, INT_STAT, DELAY.	Covers observation of reset values.	R2

9. SVA Assertion Inventory

spi_sva.sv contains APB/regfile and SPI-core assertion modules. The assertions are bound through tb_top.u_wrap.u_dut.u_regfile and tb_top.u_wrap.u_dut.u_core. Assertion failures print [ASSERTION_ERROR] so the grading logs can identify bug detections.

Register-file and APB assertions

Assertion	Area	Req	Purpose
APB_1_SETUP_THEN_ACCESS	APB	R22	Checks that each APB SETUP phase is followed by a valid ACCESS phase.
APB_2_PENABLE_NEEDS_PSEL	APB	R22	Ensures PENABLE is never asserted unless PSEL is asserted.
APB_3a_ADDR_CTRL_STABLE	APB	R22	Ensures PADDR and PWRITE remain stable from SETUP to ACCESS.
APB_3b_WDATA_STABLE_ON_WRITE	APB	R22	Ensures PWDATA remains stable from SETUP to ACCESS for writes.
APB_4a_PRDATA_RESERVED_ZERO	APB	R23	Checks that reserved offset reads return 32h0 during ACCESS.
APB_4b_PRDATA_WO_REG_ZERO	APB	TX_DATA read-zero behavior	Checks that reads from write-only TX_DATA return 32h0.
APB_5_PREADY_ALWAYS_1	APB	R22	Requires PREADY=1 during every ACCESS phase.
APB_6_PSLVERR_ALWAYS_0	APB	R22	Checks PSLVERR remains deasserted.

IRQ_1_TX_OVF_SET_ON_OVERFLOW	IRQ/FIFO	R13	Checks that a dropped TX push due to full FIFO sets INT_STAT[TX_OVF].
IRQ_2_IRQ_EQUATION	IRQ	R16	Checks IRQ equals reduction OR of INT_STAT & INT_EN.
IRQ_3a_TX_EMPTY_STICKY	IRQ	R17	Checks TX_EMPTY remains sticky unless W1C clears it.
IRQ_3b_RX_FULL_STICKY	IRQ	R17	Checks RX_FULL remains sticky unless W1C clears it.
IRQ_3c_TX_OVF_STICKY	IRQ	R17	Checks TX_OVF remains sticky unless W1C clears it.
IRQ_3d_RX_OVF_STICKY	IRQ	R17	Checks RX_OVF remains sticky unless W1C clears it.
IRQ_3e_DONE_STICKY	IRQ	R17	Checks TRANSFER_DONE remains sticky unless W1C clears it.
IRQ_4_INT_STAT_ZERO_AFTER_RESET	IRQ/Reset	R2	Checks INT_STAT is zero after reset.
FIFO_2_RX_COUNT_BOUNDED	FIFO	R12	Checks RX FIFO count never exceeds depth 8.
FIFO_3_RX_OVF_SET_ON_OVERFLOW	FIFO/IRQ	R14	Checks RX overflow sets INT_STAT[RX_OVF].

SPI-core assertions

Assertion	Area	Req	Purpose
SPI_1_SCLK_IDLE_POLARITY	SPI	R4	Checks SCLK equals CPOL when core is idle and CPOL is stable.
SPI_2A_MOSI_STABLE_RISING_SAMPLE	SPI	R5	Checks MOSI stability on rising sample edges for modes whose latched CPOL equals CPHA.
SPI_2B_MOSI_STABLE_FALLING_SAMPLE	SPI	R5	Checks MOSI stability on falling sample edges for modes whose latched CPOL differs from CPHA.
SPI_3_START_REQUIRES_SS_ASSERTED	SPI	R20 / transfer precondition	Checks transfer does not start while all slave selects are deasserted.
SPI_5_TX_POP_ONE_CYCLE_PULSE	SPI/FIFO	R9	Checks tx_pop is a one-PCLK pulse.
SPI_6_DONE_PULSE_ONE_CYCLE	SPI/IRQ	R7, R16	Checks transfer_done_pulse is a one-PCLK pulse.

10. Scoreboard and Checker Plan

- Register checker: predicts reset values, R/W masking, write-only TX_DATA read-zero behavior, reserved offset read-zero behavior and W1C effects on INT_STAT.
- FIFO checker: maintains TX/RX queues, verifies FIFO order, depth-8 full/empty behavior, overflow discard behavior and RX_DATA pop behavior.

- Interrupt checker: predicts sticky INT_STAT bits, masked-event status capture, INT_EN gating, W1C clearing and W1C/event race behavior where the event wins.
- SPI data checker: predicts TX/RX words using width, MSB/LSB bit order, loopback enable and the SPI slave MISO pattern; loopback expects RX to equal TX and external MISO to be ignored.
- Clock-divider checker: measures PCLK cycles between SCLK toggles. Expected half-period is DIV+1 PCLK cycles and full period is 2*(DIV+1) PCLK cycles.
- Delay checker: verifies consecutive transfers with programmed DELAY, including BUSY staying active through the inter-transfer gap where applicable.
- Error reporting: all detected mismatches increment error_count and print [SCOREBOARD_ERROR], while assertions print [ASSERTION_ERROR].

11. Regression and Coverage Closure Plan

Item	Plan
Mandatory tests	Regression includes the ten required test names exactly as specified by the grading contract.
All-test mode	tb_top.sv supports TESTNAME=all to execute the implemented test list sequentially.
Seeds	Use +SEED for deterministic randomized tests and keep total simulation invocations below 10000.
Coverage target	Functional and code coverage targets are at least 85 percent on golden RTL according to the grading contract.
Coverage closure	Use coverage.sv bins to identify gaps and adjust tests or stimulus constraints.
Timeout	tb_top.sv includes a global timeout to avoid hung simulations.
Compile command	make clean; make compile
Single-test command	make run TEST=sanity_test SEED=1
Regression command	make regress REGRESSION_SEEDS=<N>
Parallel runner	python run_parallel.py --seeds <N> --jobs <J>
Coverage command	make cov

12. References

- SPI Master Controller Specification, Rev 1.1, Spring 2026.
- Grading Interface Contract - SPI Master Verification Project.
- Project README - Final Project SPI Master Controller Verification.
- Final SV project files: tb_top.sv, Makefile, run_parallel.py, coverage.sv, ref_model.sv, spi_sva.sv, stim.sv, BFM's and the ten mandatory tests.